

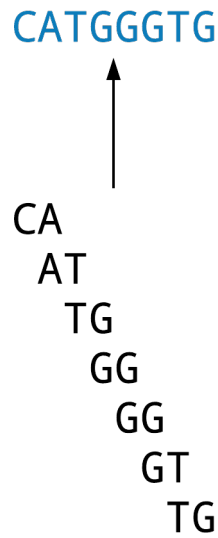
3B Reconstruct a String from its Genome Path

String Spelled by a Genome Path Problem

Find the string spelled by a genome path.

Input: A sequence of k -mers $Pattern_1, \dots, Pattern_n$ such that the last $k - 1$ symbols of $Pattern_i$ are equal to the first $k - 1$ symbols of $Pattern_{i+1}$ for i from 1 to $n - 1$.

Output: A string $Text$ of length $k + n - 1$ where the i -th k -mer in $Text$ is equal to $Pattern_i$ for all i from 1 to n .



Formatting

Input: A space-separated list of strings $Patterns$.

Output: A string $Text$.

Constraints

- The number of patterns in the string-set $Pattern$ will be between 1 and 10^4 .
- The length of any one pattern in $Pattern$ will be between 1 and 10^2 .

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:

ACCGA CCGAA CGAAG GAAGC AAGCT

Output:

ACCGAAGCT

Case 2

Description: The sample dataset is not actually run on your code.

Input:

CTT TTT TTG

Output:

CTTTG

Case 3

Description: The sample dataset is not actually run on your code.

Input:

TT TG GT TT

Output:

TTGTT

Case 4

Description: The sample dataset is not actually run on your code.

Input:

AGCAGATC GCAGATCA CAGATCAT AGATCATC GATCATCG ATCATCGG

Output:

AGCAGATCATCGG

Case 5

Description: A larger dataset of the same size as that provided by the randomized autograder.